

① $\deg(p(x)) = 1 \Rightarrow p(x)$ 不可约

∵ 若 $p(x)$ 可约 则 $p(x) = a(x) \cdot b(x)$

其中 $\deg(a(x))$ 和 $\deg(b(x)) < 1$

矛盾 ∴ $p(x)$ 不可约

② 设 $(p(x), f(x)) = g(x)$ 则 $\exists a(x), b(x)$

则 $p(x) = g(x) \cdot a(x)$ $f(x) = g(x) \cdot b(x)$

又 ∵ $p(x)$ 不可约, ∴ $\deg g(x) = \deg(p(x))$

或 $\deg(g(x)) = 1$ 前者可知 $g(x) = c p(x)$

后者知 $g(x) = c$ 其中 c 为常数

∴ $(p(x), f(x)) = 1$ 或 $(p(x), f(x)) = c p(x)$

③ $p(x), p'(x)$ 均为不可约多项式

且 $\nexists c, p'(x) = cp(x)$

由②可知, $(p(x), p'(x)) = 1$ 或 $(p(x), p'(x)) = cp(x)$

$\because p'(x) \nmid cp(x) \therefore (p(x), p'(x)) = 1$

④ 若 $p(x)$ 为 $f(x)$ k 重因式

$$f(x) = p^k(x) g(x) \quad p(x) \nmid g(x)$$

$$f'(x) = p^{k-1}(x) (kg(x) \cdot p'(x) + p(x) g'(x))$$

$\because k \geq 1 \therefore p(x) \mid p^{k-1}(x) \therefore p(x) \mid f(x)$ 且 $p(x) \mid f'(x)$

$\therefore p(x) \mid (f(x), f'(x))$

反过来, 若 $p(x) \mid (f(x), f'(x))$

$$\text{则 } f(x) = p^{s_1}(x) \cdot g(x) \quad f'(x) = p^{s_2}(x) \cdot h(x) \quad p(x) \nmid h(x)$$

$$f'(x) = p^{s_1-1}(x) (s_1 k \cdot g(x) \cdot p'(x) + p(x) g'(x))$$

$\hookrightarrow p(x) \nmid s_1 k g(x) p'(x)$

$\therefore p^{s_1-1}(x) = p^{s_2}(x) \quad \because s_2 \geq 1 \therefore s_1 \geq 2 \therefore p \text{ 为 } f(x) \text{ 重因式}$

⑤ 由④知 $p(x)$ 为 $f(x)$ 重因式 $\Leftrightarrow p(x) \mid (f(x), f'(x))$

若 $(f(x), f'(x)) = 1$ 则 $\forall p(x) \mid 1$ 则 $p(x) = c$

c 为常数 $\therefore f(x)$ 无重因式

若 $f(x)$ 无重因式, $f(x) = p(x) g(x)$ $p(x) \nmid g(x)$

$$f'(x) = p'(x)g(x) + g'(x)p(x)$$

$\because \deg(p(x)) = 1 \therefore p(x)$ 不可约

且 $p'(x) = c$ c 为常数 $(p(x), p'(x)) = 1$

又 $\because p(x) \nmid g(x) \therefore p(x) \nmid p'(x)g(x)$

$\therefore p(x) \nmid f'(x)$ $\therefore \forall p(x)$ 不为 $f'(x)$ 因子
($p(x), f'(x)$) = 1

$f(x) = \prod_i^n p_i(x)$ $\therefore$ 可知 $(f(x), f'(x)) = 1$